FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – II
PAPER –1
TEST DATE: 11-12-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –06, 19 – 24, 37 – 42):* This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(07 – 10, 25 – 28, 43 – 46):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(11 – 18, 29 – 36, 47 – 54):* This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

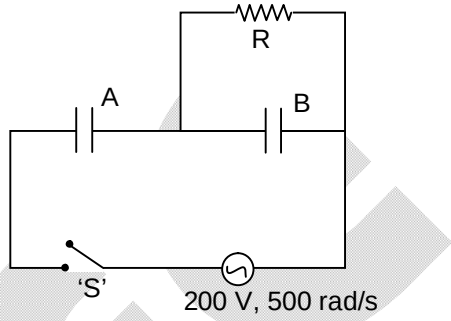
Physics

PART – I

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

1. A resistor 'R' of resistance 2000 Ω and two similar capacitor 'A' and 'B' of capacitance 1 μC each are connected with an A.C. source of 200 Volt and 500 rad/s. Now switch is closed at t = 0, then choose the correct option(s).



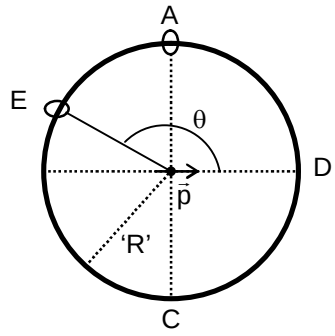
(A) The phase difference between applied voltage and voltage across the capacitor A is $\tan^{-1}\left(\frac{1}{3}\right)$

(B) The phase difference between current supplied by source and current in the capacitor 'B' is zero.

(C) The current in the capacitor 'A' is $\frac{1}{5\sqrt{20}}$ ampere.

(D) Power dissipated through resistor is 4 watt.

2. A ring of radius R is fixed and kept in a vertical plane. A small dipole of dipole moment $\vec{p}$ is fixed at the centre of ring. A small bead of mass 'm' and charge q is released from point A. Bead can slide on the ring. The inner diameter of bead and thickness of ring are nearly equal. Ring is insulating and frictionless. Ignore effect of gravity. Now choose the correct option(s).



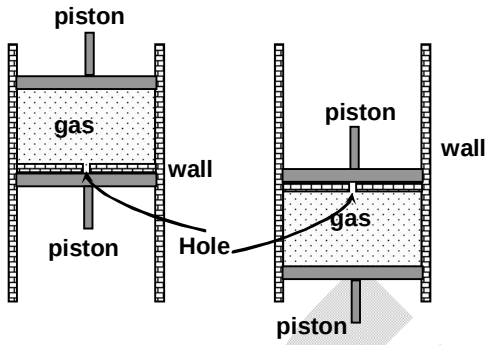
(A) Bead will never reach at point D.

(B) The normal force between bead and ring at 'E' is directly proportional to the charge on the bead.

(C) The motion of bead is periodic.

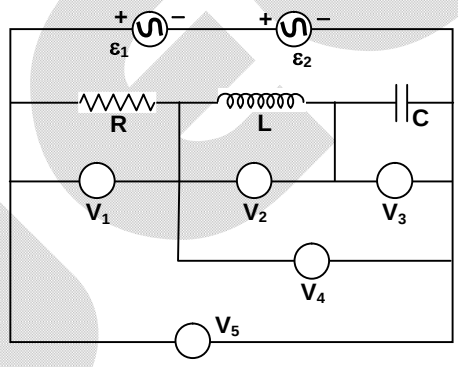
(D) Speed of bead at point 'E' is given by $\sqrt{\frac{-p(\cos\theta)q}{2\pi\varepsilon_0 mr^2}}$

3. There is a fixed long vertical cylinder of good heat insulator with open ends, is separated from its surroundings by two insulated identical, frictionless pistons. An insulated separating wall is fixed between pistons. A very small hole is made at the middle of separating wall. An ideal gas is enclosed between the pistons at temperature of surrounding. Initially the lower piston is held up against the separating wall, when lower piston is released, it moves downward very slowly. Now choose the correct option(s).



(A) During the process the pressure of gas above the separating wall remains constant.
(B) During the process the pressure of gas below the separating wall remains constant.
(C) Initial and final temperature of gas remain same.
(D) Initial and final temperature of gas is different.

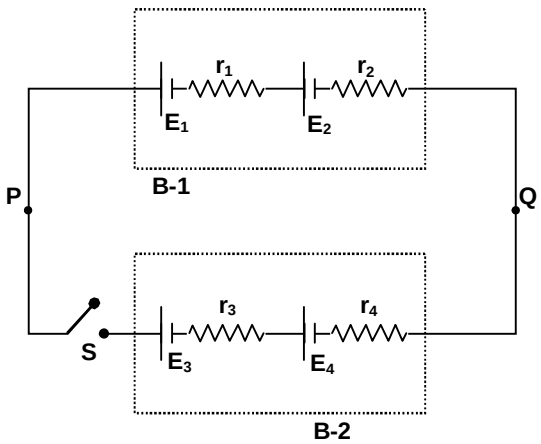
4. A resistor R = 500 Ω, an ideal inductor L = 0.9 H and a capacitor C = 2.5 μF are connected with two alternating source e_1 = 100 sin(1000t) and $e_2 = 100\sqrt{3}\cos(1000t)$. V_1, V_2, V_3, V_4 and V_5 are five ideal voltmeters and connected in the circuit as shown in the figure. Choose the correct option(s).



(A) Phase difference between voltage of V_3 and V_5 is $\frac{\pi}{2}+\tan^{-1}\left(\frac{9}{5}\right)$

(B) Reading of V_4 is 180 volt
(C) Difference between reading of V_2 and V_1 is 80 volt.
(D) Ratio of reading of V_4 and V_5 is $\frac{1}{\sqrt{2}}$

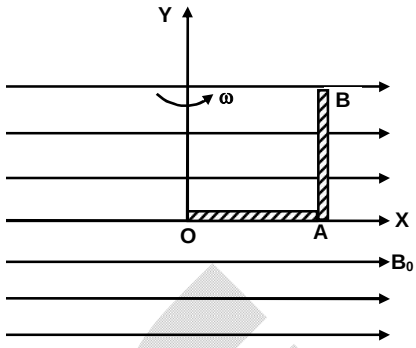
5. A battery B-1 is made by using two cells of emf E_1 and E_2 and internal resistance r_1 and r_2 respectively. Another battery B-2 is made by using different cells of emf E_3 and E_4 and internal resistance r_3 and r_4 respectively. Now they are connected with points P and Q as shown in the figure. All cells are rechargeable. Now switch is closed at t = 0, then choose the correct option(s).



(A) The potential difference across both the battery will be equal.
(B) Rate of heat dissipation in both batteries is equal.
(C) One battery will charge the other battery.
(D) Rate of loss in chemical energy of one battery is more than the rate of gain in chemical energy of the other.

6. A L-shaped conducting rod OAB is rotating in a constant magnetic field of B_0 = 100 Tesla acting along x-axis. The length of rod OA = 20 cm and AB = 20 cm. If rod is rotating about y-axis with constant angular velocity of 5 rad/second then at a given instant.

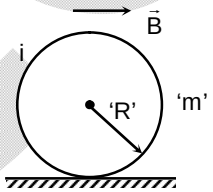
(A) Potential difference across OA is 10 Volt
(B) Potential difference across OA is zero Volt
(C) Potential difference across AB is 20 Volt
(D) Potential difference across OB is 10 Volt



Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

7. A ring of mass m = 10 kg and radius R = 5 cm is kept at rest on a smooth horizontal surface in vertical plane as shown in the figure. The ring carries a current i = 7 ampere. A constant horizontal magnetic field B = 100 mT is switched on at t = 0.



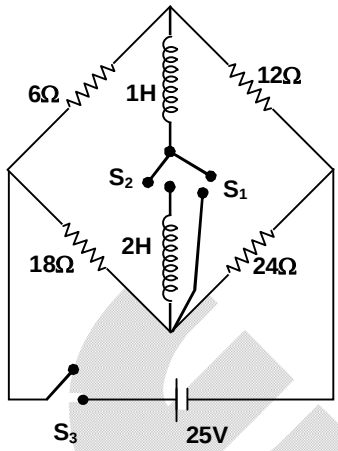
List –I describes the statement regarding ring and List-II gives their respective values.

List –I		List –II	
(I)	The magnetic moment of loop in S.I. units	(P)	0.0880
(II)	Torque acting on loop due to magnetic field in S.I. units at t = 0	(Q)	0.0044
(III)	Angular acceleration of loop in S.I. units at t = 0	(R)	0.0055
(IV)	Kinetic energy of ring when it has rotated through 90°	(S)	0.4400
		(T)	0.0550

Which one of the following options is correct ?

(A) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ R
(B) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S
(C) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ P, IV $\rightarrow$ Q
(D) I $\rightarrow$ T, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ R

8. Four resistors 6Ω, 12Ω, 18Ω and 24Ω, two ideal inductors 1H and 2H and a battery of 25 V are arranged in the given circuit with the help of three switches S_1, S_2 and S_3 as shown in the figure. ($e^{-1} = 0.37$).



List –I describes statement regarding the given circuit and List-II gives their values. (either exactly or as a close approximation).

List –I		List -II	
(I)	When switch S_1 and S_3 closed simultaneity and S_2 remain opened. Then time constant of circuit in second.	(P)	1.35
(II)	When switch S_1 and S_3 remain closed for long time and S_2 remain opened then current through inductor in ampere at one time constant.	(Q)	0.07
(III)	When switch S_1 and S_3 remain closed for long time and S_2 remain opened then current in 12Ω resistance at one time constant.	(R)	0.056
(IV)	If S_1 and S_3 remain closed for long time and then S_1 is opened, S_2 is closed and S_3 remain closed then find current in the branch of inductor immediately after switching.	(S)	0.105
		(T)	0.167

Which one of the following options is correct ?

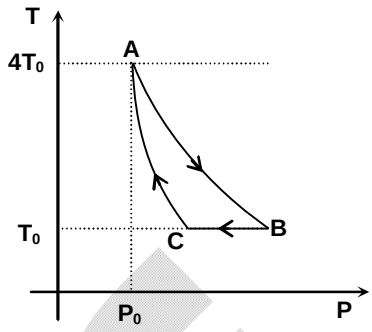
(A) I $\rightarrow$ Q, II $\rightarrow$ T, III $\rightarrow$ S, IV $\rightarrow$ R

(B) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ R

(C) I $\rightarrow$ R, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ S

(D) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ S

9. One mole of an ideal diatomic gas undergoes a cyclic process as shown in Temperature-Pressure graph. Temperature and pressure of gas at 'A' is $4T_0$ and P_0 respectively and temperature in process BC is T_0. The process AB is given by TP^2 = constant and the process CA is given by TP^4 = constant.



List –I describe the name of thermodynamics process involve in the cyclic process and List-II gives work done by gas in the process, charge in internal energy of the gas in the process and heat given to the gas in the process.

List –I		List –II	
(I)	In the thermodynamics process A $\rightarrow$ B	(P)	Work done by gas and heat given to gas is negative
(II)	In the thermodynamics process B $\rightarrow$ C	(Q)	Work done by gas is negative but heat given to gas is positive
(III)	In the thermodynamics process C $\rightarrow$ A	(R)	Work done by gas is positive but heat given to gas negative
(IV)	In the cyclic process	(S)	Work done by gas and heat given to gas is positive
		(T)	Work done by gas is positive and change in internal energy of gas is negative

Which one of the following options is correct ?

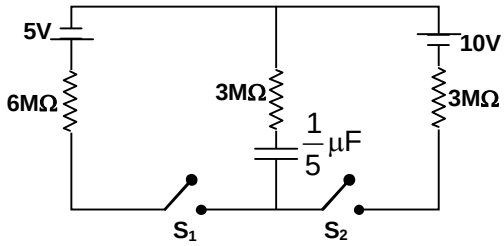
(A) I $\rightarrow$ P, II $\rightarrow$ T, III $\rightarrow$ S, IV $\rightarrow$ Q

(B) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ R

(C) I $\rightarrow$ P, II $\rightarrow$ S, III $\rightarrow$ S, IV $\rightarrow$ P

(D) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ T

10. Three resistors of resistance 6MΩ, 3MΩ and 3MΩ, one capacitor of capacitance $\frac{1}{5}\mu F$, two cells of potential difference 5V and 10V and two switches are connected in the circuit as shown in the figure. Both switches is closed simultaneously at t = 0.
($e^{-1} = 0.37$)



List-I describe the charges and currents in the circuit at a particular time and List-II gives possible values of charges and current.

List –I		List -II	
(I)	The maximum charge on the capacitor in μC	(P)	0.63
(II)	Charge on capacitor at t = 1 sec in μC	(Q)	1.54
(III)	Current in the branch of capacitor at t = 1 sec in μA	(R)	2.06
(IV)	Current passing through battery of 5V at t = 1 sec in μA	(S)	1.00
		(T)	0.37

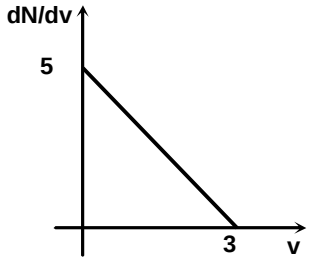
Which one of the following options is correct ?

(A) I → P, II → T, III → R, IV → S
(B) I → T, II → P, III → Q, IV → R
(C) I → P, II → R, III → T, IV → Q
(D) I → S, II → P, III → T, IV → Q

Section – B (Maximum Marks: 24)

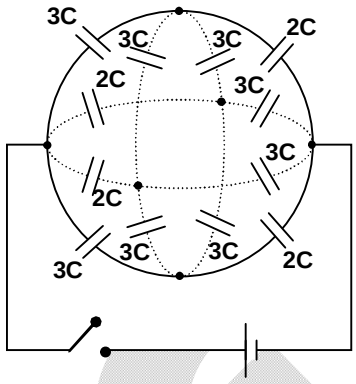
*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

11. Consider a gas kept in a closed container. 'N' is the no of molecules present in the container and v is the speed of molecules. Molecular speed distribution graph is shown in the figure with suitable S.I. units of $\frac{dN}{dv}$ and v. Find v_{rms}^2 of molecules in S.I. units.

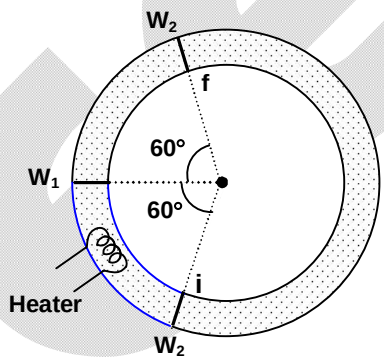


12. A semi-cylindrical shell (A) of diameter 15 cm and length $\frac{70}{22}$ meter is kept in another semi-cylindrical shell (B) of diameter 30 cm and same length such that the axis of both cylinder coincide with each other. A liquid of resistivity 100 Ω m is filled in the space between them. The lower surface of smaller cylinder A is maintained at higher potential than the larger cylinder B. Find the electrical resistance between the surface of cylinder in Ω.

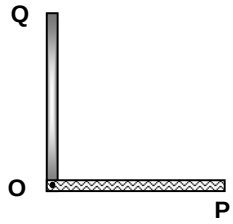
13. 12 capacitors are connected with a battery of potential difference V_0 as shown in the figure. Initially switch is open. Find total heat (in μJ) generated in the system after the switch is closed. (Given $C = \frac{1}{340}\mu F$, V_0 = 17 volt)



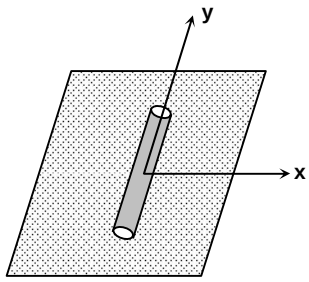
14. An insulated thin tube is fixed on horizontal surface. The total volume of tube is V_0. Inside the tube, W_1 is fixed adiabatic wall and W_2 is movable adiabatic wall. An ideal gas of adiabatic coefficient $\gamma = \frac{3}{2}$ is filled to entire volume of tube in both chambers. The initial pressure in both chamber is P_0 and number of moles of gas in both chamber are same. A heater is fixed which supplies heat very slowly such that wall W_2 is moved from position i to f, at this position wall W_2 stays in equilibrium position even the heater is switched off. Neglect friction and any loss of heat to the surrounding. The total heat supplied by heater is $\mu P_0 V_0$. Find the value of μ. ($\sqrt{5} = 2.23$)



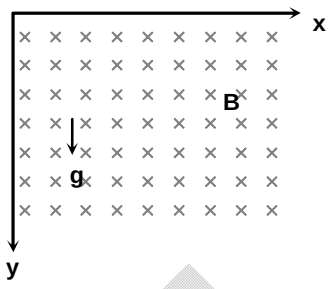
15. A L-shaped rod is made by using two different rods PO and QO. The length of rods PO and QO is 6.0 cm and 4.0 cm respectively and their coefficient of linear expansion is 1.2×10^{-5}/°C and 3.0×10^{-5}/°C respectively. The coefficient of linear expansion of the rod to be joined between point P and Q is $\lambda \times 10^5$ /°C. Find the value of λ so that no bending takes place anywhere.



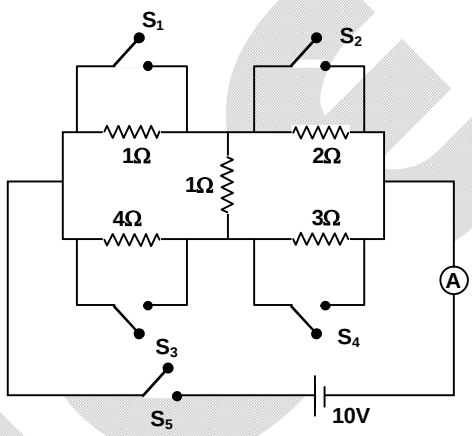
16. A uniform magnetic field B = 4 Tesla exist in the horizontal direction along x-axis. A conducting wire of mass 600 gm and length 5 cm is kept along y-axis on a rough horizontal surface. Conducting wire is carrying a current of magnitude 5 amp along y-axis. The minimum force required along x-axis to move the wire is F_1. Now the direction of current in the wire is reversed then minimum force required along x-axis to move the wire is F_2. Find the ratio of $\frac{F_1}{F_2}$.



17. A constant horizontal uniform magnetic field $B = 5 \times 10^6$ Tesla exist in a uniform vertical gravitational field $g = 10$ m/s^2. Both fields are perpendicular to each other. A particle of mass m = 50 gm and charge $q = 2\ \mu C$ is released in the space. Find the displacement (in cm) of particle along y-axis when velocity of particle is along x-axis first time.



18. Five resistors of resistance 1 Ω, 1Ω, 2Ω, 3Ω and 4Ω, a battery of voltage 10V, an ideal ammeter and five switches S_1, S_2, S_3, S_4 and S_5 are connected in a circuit as shown in the figure. When S_3 and S_5 are closed and S_1, S_2 and S_4 remain opened then current read by ammeter is i_1. When S_1, S_3 and S_5 are closed and S_2, S_4 remain opened then current read by ammeter is i_2. The ratio of current $\frac{i_1}{i_2}$ is given by 11λ. Find the value of λ.



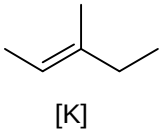
Chemistry

PART – II

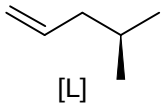
Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

19. Choose the INCORRECT statement(s) about the following alkenes is/are



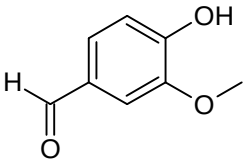
[K]



[L]

(A) Alkene [K] is more stable than alkene [L].
(B) Hydrogenation of achiral alkene [K] will give chiral product.
(C) Hydrogenation of achiral alkene [L] will give chiral product.
(D) The two alkenes [K] and [L] are structural isomers.

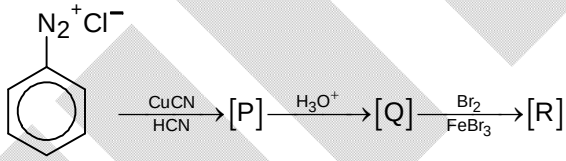
20. Choose the CORRECT statement (s) regarding Vanilin



Vanilin [P]

(A) [P] can be obtained as a major product when 3,4-dihydroxy benzaldehyde is treated with one equivalent of NaOH and one equivalent of MeBr.
(B) [P] can be obtained as a one of the product when 2-methoxy phenol is treated with chloroform and aq. KOH.
(C) Position isomer of [P] can be obtained when 3,4-dihydroxy benzaldehyde is treated with one equivalent of NaOH and one equivalent of MeBr.
(D) [P] can be obtained when 3,4-dihydroxy benzaldehyde is treated with two equivalent of NaOH and one equivalent of MeBr followed by acidic work up.

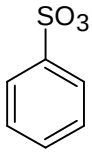
21. In the following reaction sequence:

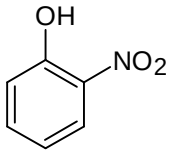


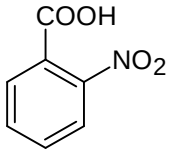
Choose the 'FALSE' statement(s):
(A) [R] is more acidic than [Q].
(B) [R] is more reactive towards electrophilic aromatic substitution than [Q].
(C) [Q] is more reactive towards electrophilic substitution than benzene.
(D) [R] on reaction with sodalime form a compound which can undergo phenol formation with aq. NaOH treatment under the normal condition of pressure and temperature.

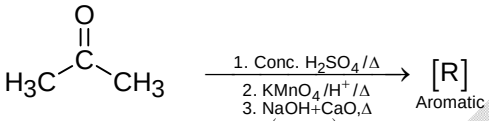
22. Which of the following compounds are soluble in $NaHCO_3$?

(A) Picric acid

(B) 

(C) 

(D) 

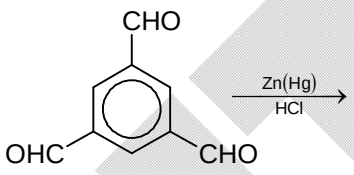
23. 

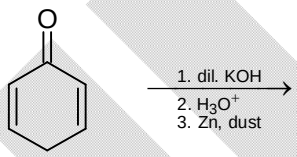
3 moles of Acetone

Out of the given reactions, which can give major product [R]?

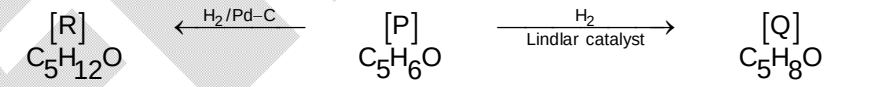
(A) $HC\equiv CH \xrightarrow[\text{Fe tube}]{\text{Red hot}}$

(B) $\text{n} - \text{hexane} \xrightarrow[\text{773 K, 10–20 atm}]{Cr_2O_3 - Al_2O_3}$

(C) 

(D) 

24. Choose the correct statements regarding the given scheme



Optically inactive Optically active Optically inactive

(A) [R] can be oxidized with PCC into optically active carbonyl compound.

(B) Oxidised product of [Q] with PCC can show tautomerism.

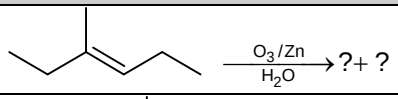
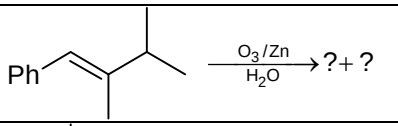
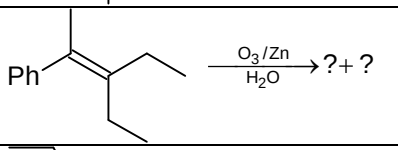
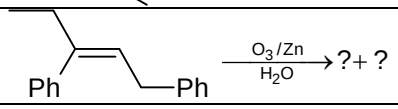
(C) [P] contains triple bond at 4th carbon (as per IUPAC)

(D) [Q] can give white turbidity immediately with Lucas test.

Section – A (Maximum Marks: 12)

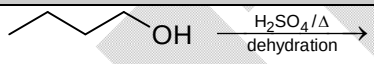
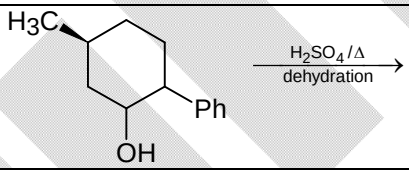
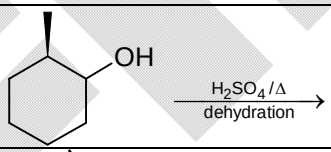
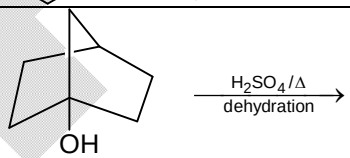
*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match each species in List – I with their properties in List-II and choose correct option

List – I		List – II (Products can be distinguished)	
(I)		(P)	Fehling's test
(II)		(Q)	Tollen's test
(III)		(R)	Iodoform test
(IV)		(S)	Schiff's test
		(T)	Benedict's test

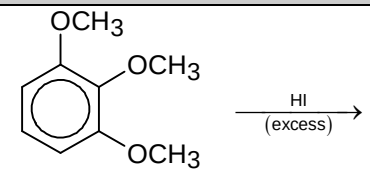
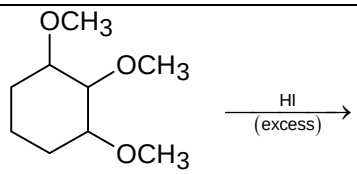
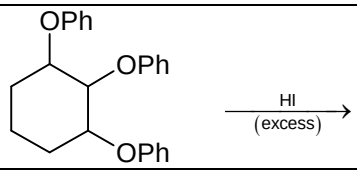
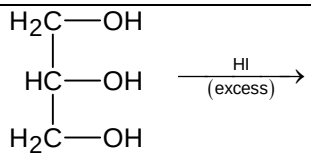
(A) I→P,Q, R, S; II→P, Q, R; III→R,S; IV→P, Q, S, T
(B) I→P, Q, R, S, T; II→Q, R, S; III→R; IV→P, Q, S, T
(C) I→Q, R, S, T; II→Q, R; III→ S; IV→P, Q, S
(D) I→R, S, T; II→P, R; III→Q,S; IV→ Q, S, T

26. Match each species in List – I with appropriate statement (s) of List – II and choose correct option

List – I		List – II	
(I)		(P)	At least one of the product shows optical isomerism.
(II)		(Q)	Major product contains even number of α – hydrogens (Hydrogens which can involve in hyperconjugation).
(III)		(R)	One of the product shows geometrical isomerism.
(IV)		(S)	Reactant can turn acidified dichromate to green.
		(T)	Will not undergo dehydration.

(A) I→Q, R; II→Q, S, T; III→P, S; IV→T
(B) I→Q, R, S; II→Q, R,S; III→P, Q; IV→R,S
(C) I→P, Q,R, S; II→Q, R, S; III→P, R; IV→Q
(D) I→Q,R, S; II→P, Q, R, S; III→P,S; IV→T

27. Match each species in List – I with appropriate statement(s) in List – II and choose correct options:

List – I		List – II	
(I)		(P)	More than 4 moles of HI will be consumed.
(II)		(Q)	Product will react with Na/dry ether.
(III)		(R)	Product formed contains 2° halide.
(IV)		(S)	CH_3I is one of the product.
		(T)	2-Iodopropane is a major product.

(A) I→Q, R; II→P, Q, R; III→P,Q,R; IV→Q, S, T
(B) I→P, T; II→Q, R,S; III→P, Q, R, T; IV→P, Q, R
(C) I→P,Q; II→Q,S, T; III→P. Q, R; IV→Q, R, T
(D) I→Q,S; II→P, Q, R, S; III→P, Q,R; IV→P, Q, R, T

28. Match each species in List – I with List – II and choose correct option

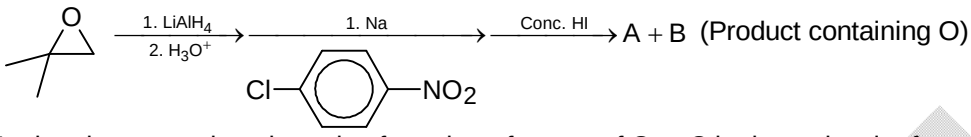
List – I		List – II	
(I)	A carbohydrate which yields only glucose on hydrolysis	(P)	Sucrose
(II)	A carbohydrate which yields glucose and fructose on hydrolysis	(Q)	Lactose
(III)	A carbohydrate which yields glucose and galactose on hydrolysis	(R)	Amylose
(IV)	A carbohydrate which can reduce Fehling's solution	(S)	Mannose
		(T)	Cellulose

(A) I→R, T; II→Q; III→P; IV→Q, S
(B) I→T; II→Q; III→Q; IV→Q
(C) I→R; II→P; III→Q; IV→P
(D) I→R, T; II→P; III→Q; IV→Q, S

Section – B (Maximum Marks: 24)

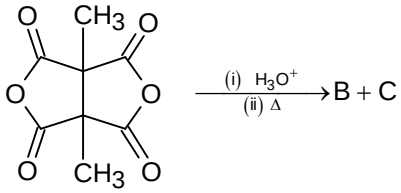
*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

29.



In the above reaction, the ratio of number of atoms of O to C in the molecular formula of B is

30.

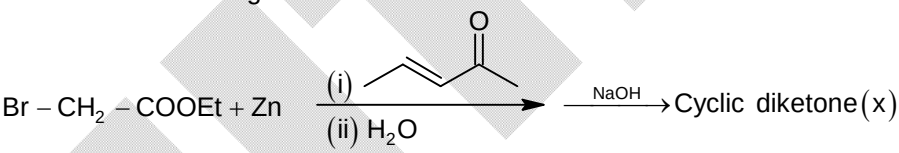


[A]

0.1 mol of (A) was reacted according to above reaction. 'C' is the gas produced which turns lime water milky. The amount of 'C' formed in gram is x. The value of $\frac{x}{2}$ is

31. 'A' is an optically active hydrocarbon with minimum molecular weight ('A' does not have any isotopic atoms like D and C^{14}).
If (a) Number of carbon atoms in A = x.
(b) Degree of unsaturation = y.
Calculate $\frac{x+y}{2}$

32. Consider the following reaction:

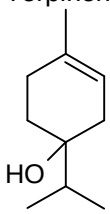


If (x) is finally treated with excess of $NaBH_4$ followed by acid work-up, the number of diols formed (including stereoisomers) is (y). Then, calculate the value of 10y.

33. How many chloroethenes (including stereoisomers) are possible for molecular formula $C_2H_{4-n}Br_n$ (where 'n' can be an integer from 1 to 4) which can possess geometrical isomerism.

34. The number of co-polymer(s) in the below list is/are X:
(Nylon-6,6, Buna-N, Dacron, Buna-S, Teflon, PHBV)
The number of carbohydrate (s) which show mutarotation in the below list is/are Y:
$\left(\begin{array}{l}\alpha-D-\text{Glucopyranose, } \beta-D-\text{Glucopyranose, Lactose,}\\ \alpha-D-\text{Fructofuranose, } \beta-D-\text{Fructofuranose, Sucrose,}\\ \text{Maltose}\end{array}\right)$
The value of $\frac{Y}{X}$?

35. Terpinen-4-ol is an active ingredient in tea tree oil has following structure



Which of the following statement(s) is/are correct about above compound?

1. It can rotate the plane polarized light.
2. It reacts with Baeyer's reagent to form triol.
3. On reaction with excess of NaBr and dilute H_2SO_4, it can form a dibromo compound.

(Note: Write your answer in three digits)

i.e. If only statement-1 is correct, then 0.01 is correct answer.
If only statement-2 is correct, then 0.02 is correct answer.
If only statement-3 is correct, then 0.03 is correct answer.
If statement-1 and 2 both are correct, then 0.12 is correct answer.
If statement-1 and 3 both are correct, then 0.13 is correct answer.
If statement-1, 2 and 3 are correct, then 1.23 is correct answer.

36. In Duma's method of estimation of nitrogen, 0.1840 g of an organic compound gave 30 ml of nitrogen collected over water at 287 K and 758 mm of Hg pressure. The percentage composition of nitrogen in the compound is…….[Given : aqueous tension at 287 K = 14 mm of Hg] [R = 0.0821 L atm mol^{-1} K^{-1}]

Mathematics

PART – III

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

37. To the parabola $x^2 = 4ay$, tangent is drawn at any point (x_1, y_1) other than the vertex. If tangents are drawn to the circle $x^2 + y^2 = 4a^2$ from any point on this tangent such that all the corresponding chords of contact pass through a fixed point (x_2, y_2), then
(A) y_1, 2a, y_2 are in Geometric Progression
(B) $-4, \frac{x_1}{x_2}, \frac{y_1}{y_2}$ are in Geometric Progression
(C) $\frac{x_1}{2}$, 2a, x_2 are in Geometric Progression
(D) $x_1x_2 + y_1y_2 = 4a^2$

38. Two ellipse with same centre and same length of major axis are such that the foci of one are on the other. If their eccentricities are e_1 and e_2, then
(A) the angle θ between their axes is given by $\theta = \sin^{-1}\left(\frac{\sqrt{(e_1^2-1)(e_2^2-1)}}{e_1e_2}\right)$
(B) quadrilateral formed by joining the foci is a parallelogram
(C) for $\theta = \frac{\pi}{4}$, $e_1^2 = \frac{2e_2^2+1}{e_2^2-2}$
(D) none of these

39. Normal to the circle $x^2 + y^2 = 9$ divides the circle having centre at $\left(3\sqrt{2}, \frac{3}{2}\right)$ and radius 3 in the areas of ratio $(2\pi - 3\sqrt{3}) : (10\pi + 3\sqrt{3})$. Then the normal can be
(A) $\sqrt{2}\,y = x$ (B) $y = \sqrt{2}\,x$
(C) $5y + \sqrt{2}\,x = 0$ (D) $\sqrt{2}\,y + x = 0$

40. If $\alpha = p\cos^3\theta \cdot \sin^2\theta$, $\beta = p\sin^3\theta \cdot \cos^2\theta$ and $\frac{(\alpha^2+\beta^2)^a}{(\alpha\beta)^b}$ (a, b $\in$ N) is independent of θ, then (a, b) is equal to
(A) (4, 5) (B) (8, 10)
(C) (5, 4) (D) (10, 8)

41. If one of the lines represented by $4x^2 + pxy + 6y^2 = 0$ is perpendicular to one of the lines given by $4x^2 + qxy - 6y^2 = 0$ and the other lines represented by them are coincident, then the value of p – 2q is
(A) –6 (B) –8
(C) 8 (D) 6

42. The expression $\frac{x^3}{2}\text{cosec}^2\left(\tan^{-1}\left(\frac{x}{y}\right)\right) + \frac{y^3}{2}\sec^2\left(\tan^{-1}\left(\frac{y}{x}\right)\right)$ is divisible by
(A) x + y (B) $x^2 + y^2$
(C) $x^3 + y^3$ (D) $x^2 - y^2$

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

43. Match the following List-I with List-II

List-I		List-II	
(I)	The number of solutions of $x^4 - 2x^2 \sin^2 \frac{\pi x}{2} + 1 = 0$ is	(P)	4
(II)	The number of solutions of $16(x^2 + 1) + \pi^2 = \lvert\tan x\rvert + 8\pi x$, $x \in \left(-\frac{\pi}{2}, \frac{3\pi}{2}\right), x \neq \frac{\pi}{2}$ is	(Q)	2
(III)	The numbers of pairs (x, y) satisfying the equation $\sin x + \sin y = \sin(x + y)$ and $\lvert x\rvert + \lvert y\rvert = 1$ is	(R)	1
(IV)	The number of integral values of x in [0, 6] satisfying $\sin x + \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x > 0$ is	(S)	3
		(T)	6

The correct option is:

(A) (I) → (T); (II) → (P); (III) → (S); (IV) → (Q)

(B) (I) → (S); (II) → (P); (III) → (R); (IV) → (Q)

(C) (I) → (Q); (II) → (P); (III) → (T); (IV) → (S)

(D) (I) → (R); (II) → (T); (III) → (P); (IV) → (S)

44. Let $t_{10} = (\sin^{-1} x)^{\sin^{-1} x} 0$, $t_2 = (\sin^{-1} x)^{\cos^{-1} x}$, $t_3 = (\cos^{-1} x)^{\sin^{-1} x}$, $t_4 = (\cos^{-1} x)^{\cos^{-1} x}$ match the following List-I with List-II

List-I		List-II	
(I)	$x \in (0, \cos 1)$	(P)	$t_1 > t_2 > t_4 > t_3$
(II)	$x \in \left(\cos 1, \frac{1}{\sqrt{2}}\right)$	(Q)	$t_4 > t_3 > t_1 > t_2$
(III)	$x \in \left(\frac{1}{\sqrt{2}}, \sin 1\right)$	(R)	$t_4 > t_1 > t_2 > t_3$
(IV)	$x \in (\sin 1, 1)$	(S)	$t_2 > t_1 > t_4 > t_3$
		(T)	$t_3 > t_4 > t_1 > t_2$

The correct option is:

(A) (I) → (T); (II) → (S); (III) → (Q); (IV) → (R)

(B) (I) → (Q); (II) → (T); (III) → (S); (IV) → (P)

(C) (I) → (R); (II) → (Q); (III) → (S); (IV) → (T)

(D) (I) → (P); (II) → (Q); (III) → (R); (IV) → (S)

45. Match the following List-I with List-II

List-I		List-II	
(I)	The shortest distance between the parabola $2y^2 = 2x - 1$ and $2x^2 = 2y - 1$ is	(P)	$\frac{2\sqrt{2}}{3}$
(II)	The triangle formed by the tangent to the parabola $y = x^2$ at the point whose abscissa is x_0 ($x_0 \in [1, 2]$), the y-axis and the straight line $y = x_0^2$ has the greatest area if x_0 is equal to	(Q)	$\frac{1}{4}$
(III)	An ellipse is inscribed in a circle and a point within the circle is chosen at random. If the probability that this point lies outside the ellipse is $\frac{2}{3}$, then the eccentricity of the ellipse is	(R)	$\frac{1}{2\sqrt{2}}$
(IV)	If $f(x) = x^3 + ax^2 + bx + c$, (a, b, c are rational numbers) and 2 roots of $f(x) = 0$ are eccentricities of a parabola and a rectangular hyperbola, then $\frac{1}{abc}$ equals	(S)	$\frac{1}{\sqrt{2}}$
		(T)	2

The correct option is:
(A) (I) → (S); (II) → (T); (III) → (Q); (IV) → (R)
(B) (I) → (P); (II) → (Q); (III) → (S); (IV) → (T)
(C) (I) → (Q); (II) → (S); (III) → (R); (IV) → (T)
(D) (I) → (R); (II) → (T); (III) → (P); (IV) → (Q)

46. Match the following List-I with List-II

List-I		List-II	
(I)	If the angle between lines belonging to the family of lines $(3x + y - 7) + \lambda(x + y - 3) = 0$ and $(x + 3y - 7) + \mu(x + 4y - 3) = 0$ which are farthest from origin (λ and μ are real parameters), is $\tan^{-1}\left(\frac{a}{b}\right)$ where a, b are coprime natural numbers, then b – a is	(P)	11
(II)	If the roots of the equation $px^2 + qx + 10 = 0$ are not real and distinct where $p, q \in R$ and α and β are the values of p and q respectively for which $5p + q$ is minimum, then the family of lines $\alpha(4x + 5\alpha y + 3) + \beta(4x + \beta y - 4) = 0$ are concurrent at (a, b). (a – b) is	(Q)	7
(III)	A circle is inscribed into a rhombus PQRS with one angle 120°. The distance from the centre of the circle to the nearest vertex is equal to 1. If A is any point on the circle, then $\|AP\|^2 + \|AQ\|^2 + \|AR\|^2 + \|AS\|^2$ is	(R)	3
(IV)	If the area bounded by the circles $x^2 + y^2 = r^2$, r = 1, 2 and the rays given by $2x^2 - 3xy - 2y^2 = 0$, $y > 0$ is $\frac{\lambda\pi}{4}$ sq. units. Then the value of λ is	(S)	1
		(T)	5

The correct option is:
(A) (I) → (P); (II) → (R); (III) → (S); (IV) → (Q)
(B) (I) → (Q); (II) → (S); (III) → (P); (IV) → (R)
(C) (I) → (T); (II) → (P); (III) → (S); (IV) → (R)
(D) (I) → (S); (II) → (P); (III) → (T); (IV) → (Q)

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

47. If $\sin^{-1}\alpha + \sin^{-1}\beta + \sin^{-1}\gamma = \pi$, then $\dfrac{\alpha\sqrt{1-\alpha^2} + \beta\sqrt{1-\beta^2} + \gamma\sqrt{1-\gamma^2}}{\alpha\beta\gamma}$ is equal to

48. If $b^2 + c^2 = 2023a^2$, where a, b, c are the side lengths of ΔABC for usual notation, then $\dfrac{\cot A}{\cot B + \cot C}$ is equal to

49. For the hyperbola $\dfrac{x^2}{25} - \dfrac{y^2}{45} = 1$, the chord x cos θ + y sin θ = λ subtends a right angle at the centre. Then the length of tangent to the circle, concentric with this hyperbola, to which the given chord is a tangent, from the point $P\left(5, 5\sqrt{2}\right)$ is

50. Two circles S and S′ pass through both the points P(1, –2) and Q(2, 3) such that the line x – 3y + 9 = 0 is a common tangent touching them at A and B respectively. If the possible coordinates of a point R are (α, β) such that PARB is a parallelogram and $\alpha = \dfrac{a}{b}$, $\beta = \dfrac{c}{d}$ (both in lowest forms) with a, b, c d $\in$ I, then |abcd| is

51. For real numbers x and y satisfying $x^2 + 4y^2 - 6x + 4y + 9 = 0$. Then the absolute difference between the maximum and minimum values of (8x – 12y) is

52. There are two families of lines in a plane y = x + i, y = –x + i, where i = {0, 1, 2, 3, 4, 5}. The number of squares having diagonal lengths of odd units, formed by these lines is

53. Points P, Q, R lie on the parabola $y^2 = 4ax$. Tangents to the parabola at these points taken in pairs, intersect at A, B, C. The ratio of the areas of ΔABC and ΔPQR is

54. The sum of all possible values of α (in degrees) such that $\cos^3 3\alpha + \cos^3 5\alpha = 8\cos^3 4\alpha \cos^3 \alpha$ where α is measured in degrees and $\alpha \in (100°, 200°)$, is